

MEIKO AUTOMATION

# MGAA-V1

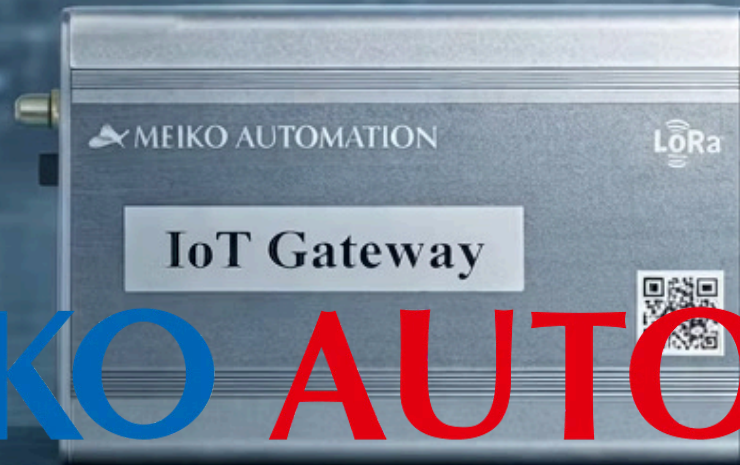
Industrial IoT Audio & Light Alert Device

Real-Time AI-Driven Warning Platform for Smart Factory

- MQTT + HTTP Protocol
- PoE Powered (802.3af/at)
- OTA Firmware Update
- AI Integration Ready



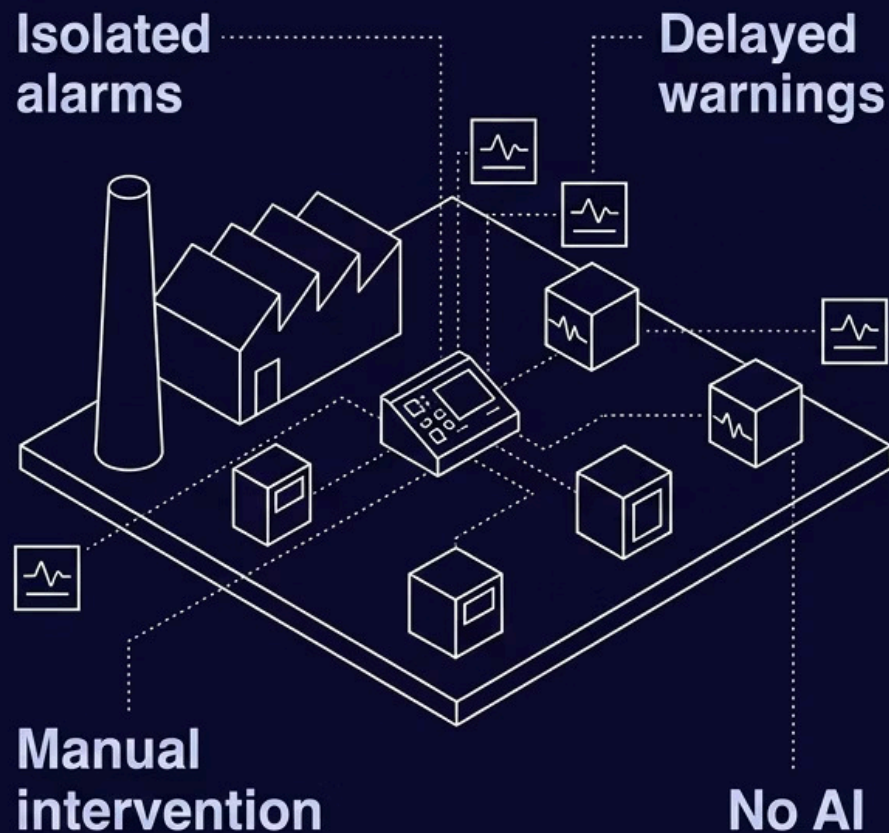
# MEIKO AUTOMATION



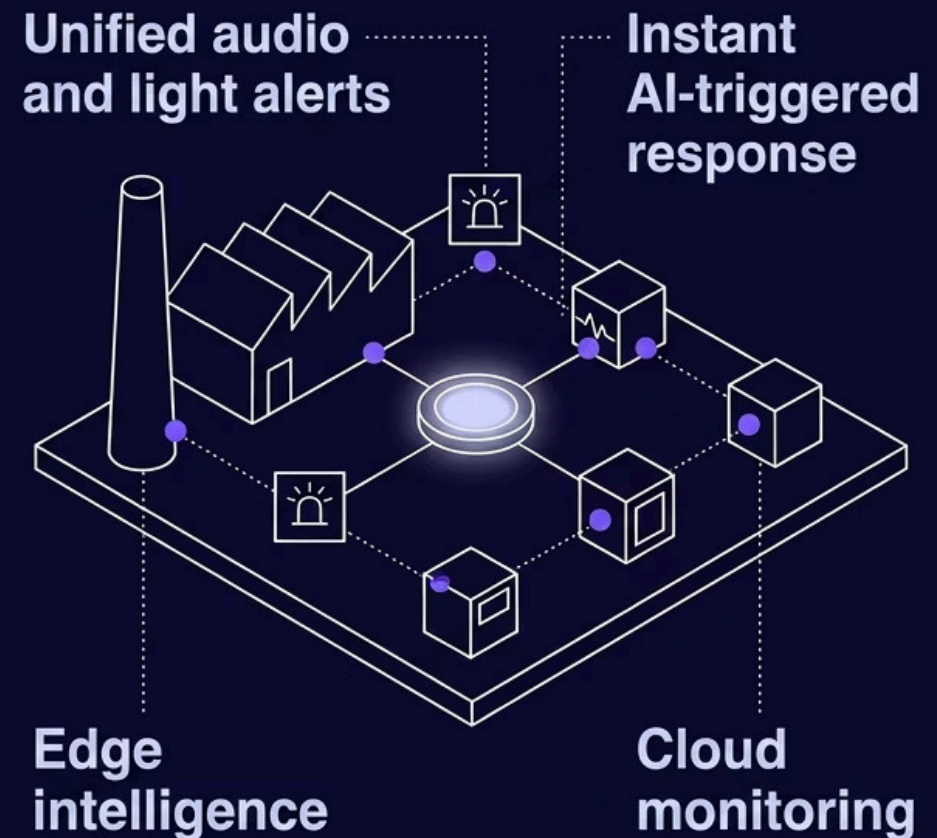


# Built for Real-Time Industrial Safety

## BEFORE: ISOLATED ALARMS



## AFTER: UNIFIED SAFETY



The MGAA-V1 transforms industrial environments by enabling **instant AI-triggered alerts**, a **unified audio and light warning system**, robust **MQTT communication**, comprehensive **cloud monitoring**, and powerful **edge intelligence** for unparalleled real-time safety.



# One Device. Complete Industrial Warning Solution.



## Audio Alert

8W Speaker, ~90dB @1m



## Visual Warning

Red/Yellow/Green Beacon



## IoT Gateway

MQTT + HTTP Connectivity



## AI Integration

Real-Time Detection Ready



## Remote Management

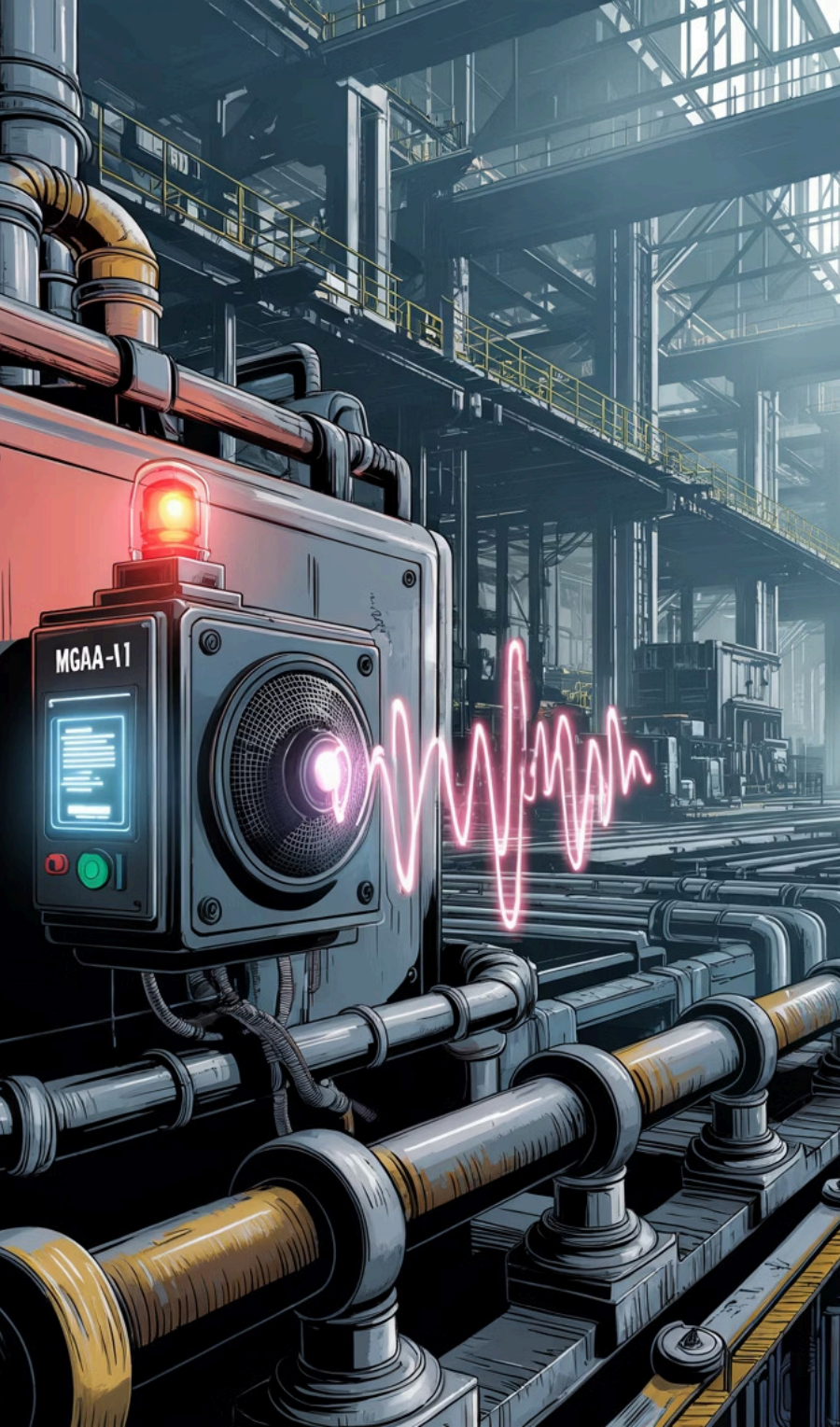
Cloud Dashboard Control



## OTA Firmware

Wireless Remote Updates



An illustration of an industrial facility with complex piping and structural elements. In the foreground, a grey industrial speaker labeled 'MGAA-11' is mounted on a pipe. It features a red beacon light on top and a glowing pink sound wave emanating from its speaker grille. The background shows a multi-level industrial structure with yellow safety railings.

# Real-Time Audio & Visual Alerting

## Powerful Audio Output

An 8W industrial speaker delivers ~90dB @1m for clear, impactful sound in noisy environments.

## Configurable Visuals

Integrated Red, Yellow, Green beacon supports ON/OFF, Blink, and Pattern modes for diverse alert signaling.

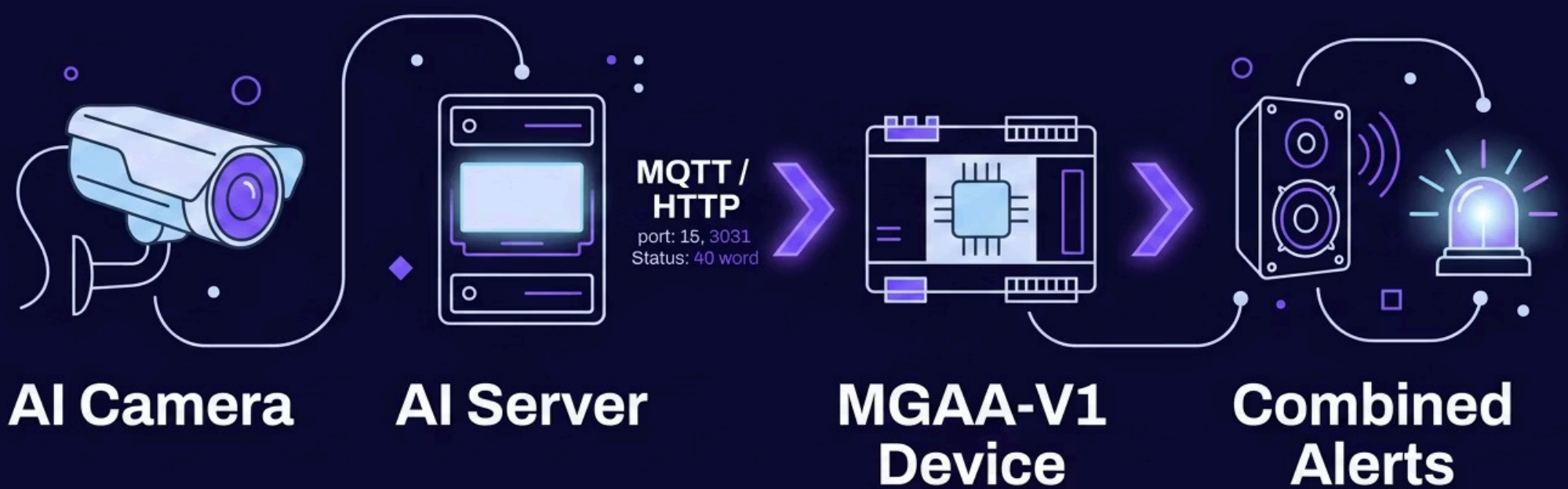
## Guaranteed Playback

Offline playback capability ensures critical audio warnings function even without network connectivity.

## Rapid Response

Achieve ultra-low latency of  $\leq 2$  seconds from trigger to alert, critical for immediate safety actions.

# Integrated with Industrial AI Systems



## MQTT Communication

Standardized, lightweight messaging for industrial IoT.



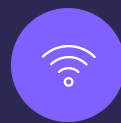
## HTTP API Support

Flexible integration with existing IT systems and applications.



## Ethernet Connectivity

Reliable wired connection for critical deployments.



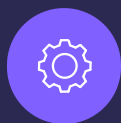
## WiFi Support

Wireless deployment flexibility where cabling is challenging.



## Modbus RS485

Seamless integration with legacy industrial equipment.



## Remote Configuration

Manage settings from anywhere, reducing on-site visits.



## OTA Firmware Updates

Future-proof your device with over-the-air improvements.

# Engineered for Continuous Industrial Operation

## PoE 802.3af/at

Flexible power input for simplified deployment.

## 12V Backup Power

Ensures continuous operation during power interruptions.

## OVP / UVP / OCP

Comprehensive Over/Under-Voltage and Over-Current Protection.

## Reverse Polarity Protection

Safeguards against incorrect power connections.

## Online/Offline Monitoring

Real-time status updates and fault detection.

## Remote Diagnostics

Troubleshoot and maintain devices without on-site visits.

## Hardware Watchdog

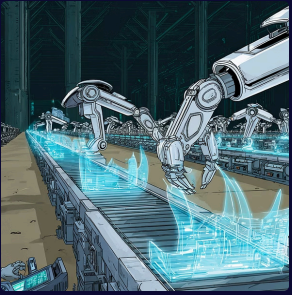
Automatic system reset for maximum uptime.

## Industrial Enclosure

Robust design for harsh factory environments.

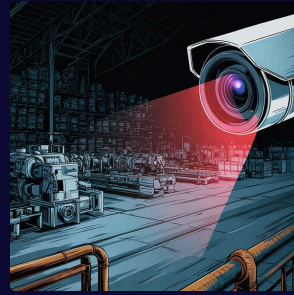


# Deployed Across Critical Industrial Environments



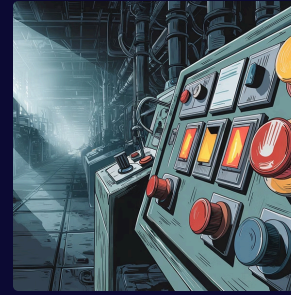
## Smart Factory

AI-driven production line monitoring



## AI Camera Warning

CCTV-AI integration for anomaly detection



## Fire Alarm Integration

Multi-zone evacuation alerts



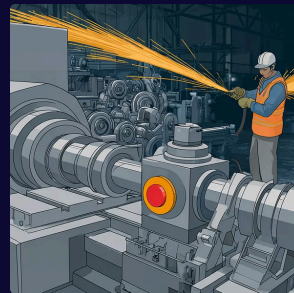
## Restricted Area Alert

Perimeter breach notification



## Warehouse Safety

Inventory and access control warnings



## Machine Safety

Equipment safeguard alerts



## Server Room Alert

Environmental monitoring and alerts



## Industrial Emergency Warning

Critical facility-wide notifications

# Technical Specifications

Engineering Datasheet — MEIKO AUTOMATION MGAA-V1

|                          |                                           |
|--------------------------|-------------------------------------------|
| Model                    | MGAA-V1                                   |
| Device Type              | Industrial IoT Audio & Light Alert Device |
| Power Input              | PoE 802.3af/at + 12V Backup               |
| Network                  | Ethernet RJ45                             |
| Wireless                 | WiFi                                      |
| Protocol                 | MQTT / HTTP API                           |
| Speaker Output           | 8W Class D                                |
| Sound Pressure           | ~90dB @1m                                 |
| Beacon                   | Red / Yellow / Green LED                  |
| Alert Modes              | ON / OFF / Blink / Pattern                |
| Modbus                   | RS485 Support                             |
| Firmware                 | OTA Supported                             |
| Monitoring               | Online / Offline Status                   |
| Over-voltage protection  | 27V                                       |
| Under-voltage protection | 7V                                        |
| Over-current protection  | 3.8A                                      |
| Reverse polarity         | Supported                                 |
| Dimensions               | 120 × 88 × 38 mm                          |
| Latency                  | ≤1–2 seconds                              |
| Enclosure                | Industrial-grade aluminum                 |
| Operating Temperature    | -10°C to +50°C                            |
| Humidity                 | 10% to 90% (non-condensing)               |



# OEM / ODM Ready Platform

Our robust platform is designed for seamless integration and customization, empowering partners with tailored solutions for diverse industrial applications.

## Custom Firmware Development

Tailor device behavior and integrate unique functionalities to meet specific project requirements.

## Private Label Branding

Offer the MGAA-V1 under your own brand with custom logos and packaging, enhancing market presence.

## Hardware Modification Capability

Adapt hardware to specific environmental or operational needs, ensuring optimal performance.

## Mass Production Support

Leverage our efficient manufacturing capabilities for large-scale deployment and timely delivery.

## System Integration Services

Expert assistance in integrating the MGAA-V1 into existing industrial ecosystems and control systems.

## Industrial Customization Options

From specialized enclosures to unique communication protocols, we provide comprehensive customization.

# Real-Time Industrial Intelligence

MEIKO AUTOMATION

Our Services:

- Industrial IoT Solutions
- Smart Factory Integration
- OEM / ODM Services
- AI Integration Support
- System Integration

Contact Information:

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Product: MGAA-V1 — Industrial IoT Audio & Light Alert Device

**Enquire about volume pricing, custom firmware development, and dedicated integration support for your deployment.**